EduFin SQL Business Challenge

Phase 1 — Monday Submission

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STEP 1A — Understand the Data Structure

```
SELECT
    loan_status AS `Loan Status`,
    COUNT(*) AS `Loan Count`,
    CONCAT(ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1), '%') AS `Percentage`
FROM loans
GROUP BY loan_status
ORDER BY `Loan Count` DESC
```

Loan Status	Loan Count	Percentage
Active	3965	79.3%
Defaulted	590	11.8%
Overdue	273	5.5%
Closed	172	3.4%

STEP 1B — Basic Portfolio Metrics

```
SELECT
    COUNT(DISTINCT customer_id) AS `Total Unique Customers`,
    COUNT(*) AS `Total Loans`,
    CONCAT(ROUND(SUM(loan_amount) / 10000000.0, 2), ' Cr') AS `Total Portfolio Value`,
    CONCAT(ROUND(AVG(loan_amount) / 100000.0, 2), ' L') AS `Average Loan Amount`
FROM loans;
```

Total Unique Customers	Total Loans	Total Portfolio Value	Average Loan Amount
3000	5000	204.82 Cr	4.1 L

STEP 1C — Calculate Risk Metrics

```
SELECT
    -- Risk Category Counts
    COUNT(CASE WHEN loan_status = 'Active' THEN 1 END) AS `Active Loans`,
```

```
    COUNT(CASE WHEN loan_status = 'Closed' THEN 1 END) AS `Closed Loans`,
    COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) AS `Defaulted Loans`,
    COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) AS `Overdue Loans`,

    -- default rate percentage %
    CONCAT(
        ROUND(
            COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*),
            2
        ),
        '%'
    ) AS `Default Rate %`,
    -- potfolio at risk percentage %
    CONCAT(
        ROUND(
            COUNT(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN 1 END) *
100.0 / COUNT(*),
            2
        ),
        '%'
    ) AS `Portfolio at Risk %`
FROM loans;
```

Active Loans	Closed Loans	Defaulted Loans	Overdue Loans	Default Rate %	Portfolio at Risk %
3965	172	590	273	11.80%	17.26%

STEP 1D — Financial Impact (Crores Format)

```
SELECT
    CONCAT(
        ROUND(SUM(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN loan_amount *
1.0 ELSE 0.0 END) / 10000000.0,
            2
        ),
        ' Cr'
    ) AS `Total Portfolio at Risk Value`
FROM loans;
```

Total Portfolio at Risk Value
35.18 Cr

STEP 1E — Executive Dashboard (Final Consolidated Query)

```
SELECT
    COUNT(*) AS `Total Loans`,
    -- For Active Loans
    COUNT(CASE WHEN loan_status = 'Active' THEN 1 END) AS `Active Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Active' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Active Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Active' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Active Portfolio Value`,
    -- For Closed Loans
    COUNT(CASE WHEN loan_status = 'Closed' THEN 1 END) AS `Closed Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Closed' THEN 1 END) * 100.0 /
```

```
COUNT(*), 1), '%') AS `Closed Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Closed' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Closed Portfolio Value`,
    -- For Defaulted Loans
    COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) AS `Defaulted Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Defaulted Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Defaulted Portfolio Value`,
    -- For Overdue Loans
    COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) AS `Overdue Loans`,
    CONCAT(ROUND(COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) * 100.0 /
COUNT(*), 1), '%') AS `Overdue Loan %`,
    CONCAT(ROUND(SUM(CASE WHEN loan_status = 'Overdue' THEN loan_amount END) /
10000000.0, 2), ' Cr') AS `Overdue Portfolio Value`,
    -- Other Metrics
    COUNT(DISTINCT customer_id) AS `Total Customers`,
    CONCAT(ROUND(SUM(loan_amount) / 10000000.0, 2), ' Cr') AS `Total Portfolio Value`,
    CONCAT(ROUND(AVG(loan_amount) / 100000.0, 2), ' L') AS `Average Loan Amount`,
    CONCAT(
        ROUND(
            COUNT(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN 1 END) *
100.0 / COUNT(*),
            2
        ),
        '%'
    ) AS `Portfolio at Risk %`,
    CONCAT(
        ROUND(
            SUM(CASE WHEN loan_status IN ('Defaulted', 'Overdue') THEN loan_amount
END) / 10000000.0,
            2
        ),
        ' Cr'
    ) AS `Total at Risk Value`,
    CASE
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 5.00 THEN 'HEALTHY'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 10.00 THEN 'MODERATE RISK'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 15.00 THEN 'HIGH RISK'
        ELSE 'CRITICAL'
    END AS `Portfolio Health Status`,
    CASE
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) > 12.00
            THEN 'Crisis team + external recovery agencies'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) > 10.00
            THEN 'Enhanced collection + review underwriting'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) > 5.00
            THEN 'Enhanced monitoring required'
        ELSE 'Standard monitoring protocols'
    END AS `Recommend Board Action`
FROM loans
```

Total Loans	Active Loans	Active Loan %	Active Portfolio Value	Closed Loans	Closed Loan %	Closed Portfolio Value	Defaulted Loans	Defaulted Loan %	Defaulted Portfolio Value	Overdue Loans	Overdue Loan %	Overdue Portfolio Value	Total Customers	Total Portfolio Value	Average Loan Amount	Portfolio at Risk %	Total at Risk Value	Portfolio Health Status	Recommend Board Action
5000	3965	79.3%	162.41 Cr	172	3.4%	7.24 Cr	590	11.8%	24.24 Cr	273	5.5%	10.94 Cr	3000	204.82 Cr	4.1 L	17.26%	35.18 Cr	HIGH RISK	Enhanced collecti...

Custom Column

```
CASE
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
5.00 THEN 'Stable'
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
8.00 THEN 'Monitor'
    WHEN COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*) <
10.00 THEN 'Alert'
```

```
    ELSE 'Emergency'
END AS risk_tier
```

> I created these thresholds to align with the current portfolio condition, where the default rate is already 11.8%, indicating systemic stress rather than normal variation. The bands are intentionally set stricter compared to standard BRD categories thresholds as it helps to identify future risks quicker, avoiding the escalation of risks, before it reaches stress levels as seen in the current situation.

WHY Gate

1. Does your "Financial Impact" (Step 1D) include only defaulted loans, or does it include "Overdue" as well? Why did you make that choice?

> My "Financial Impact" includes not only defaulted ones but also the overdue ones as well. I made that choice because overdue loans are defaults waiting to happen (basically they are at-risks as well). Also, they are potential losses which needs to be added when calculating total capital under stress consuming resources as per the instructions given in BRD.

2. If the CEO asks if this crisis is "temporary," what specific data point from your analysis would you point to as a warning sign that it's systemic?

> I would point out the 17.26% Portfolio at Risk (PAR) as a systemic warning signal because it is not just a snapshot of missed payments, it reflects a structural breakdown in repayment health across the portfolio, not isolated borrower behavior.

3. What is the single most important number in this workbook that the board should see first? Why that one?

> The single most importnat number in this workbook is "11.80% default rate", as it is the factor that indicates our entire situation as HIGH RISK against the industry benchmarks. Also, the CEO has explicitly asked us about the default rate (What's our overall default rate?) in the BRD.

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